

NMAP CHEAT SHEET

TryHackMe Nmap Room — Complete Recap

1. SCAN TYPES

Switch	Scan Type	Notes
-sT	TCP Connect	Full 3-way handshake. No sudo needed. Loud.
-sS	SYN / Stealth	Half-open. Default with sudo. Faster & stealthy.
-sU	UDP Scan	Slow (~20 min). Use --top-ports to speed up.
-sN	NULL Scan	No flags set. Firewall evasion.
-sF	FIN Scan	FIN flag only. Firewall evasion.
-sX	Xmas Scan	FIN+PSH+URG flags. Firewall evasion.
-sn	Ping Sweep	Find live hosts. No port scan.

2. PORT STATES

State	Meaning
open	Port is accepting connections
closed	Port responded with RST — no service running
filtered	Firewall blocking — no response received
open filtered	No response — could be open or firewalled (UDP/NULL/FIN/Xmas)

3. USEFUL SWITCHES

Switch	Purpose
-p <port>	Scan specific port(s). e.g. -p 80 or -p 1-1000
-p-	Scan ALL 65535 ports
-Pn	Skip ping check (use when ICMP is blocked)
-sV	Detect service versions
-O	Detect OS
-A	Aggressive: OS + version + scripts + traceroute
-T0 to -T5	Speed (T0=slowest/stealthiest, T5=fastest/loudest)
-v / -vv	Verbose output
--top-ports N	Scan top N most common ports
-oA filename	Save output in all formats
-f	Fragment packets (firewall evasion)
--mtu N	Set packet size (must be multiple of 8)
--scan-delay Nms	Delay between packets (evade time-based IDS)
--badsum	Detect firewalls using invalid checksums

4. NSE SCRIPTS

```
Scripts are stored at: /usr/share/nmap/scripts/
```

Category	What it does
safe	Won't affect the target
intrusive	May affect/disrupt the target
vuln	Scan for vulnerabilities
exploit	Try to exploit vulnerabilities
auth	Bypass authentication
brute	Brute force credentials
discovery	Query services for more info

Script Commands:

```
Run a category: nmap --script=vuln <IP>
```

```
Run specific script: nmap --script=ftp-anon <IP>
```

```
Run multiple scripts: nmap --script=smb-enum-users,smb-enum-shares <IP>
```

```
Pass arguments: nmap --script=http-put --script-args  
http-put.url='/shell.php',http-put.file='./shell.php'
```

```
Get script help: nmap --script-help ftp-anon
```

```
Search scripts: grep 'ftp' /usr/share/nmap/scripts/script.db
```

```
Update script DB: nmap --script-updatedb
```

5. FIREWALL EVASION

NULL/FIN/Xmas scans bypass firewalls that block SYN packets — they don't use the SYN flag. However modern IDS systems detect them. They don't work reliably against Windows targets as Windows responds RST to all malformed packets.

```
-Pn: Use when ICMP is blocked (Windows default firewall blocks ping)
```

```
Example: sudo nmap -sS -Pn -p 1-5000 <IP>
```

6. TCP 3-WAY HANDSHAKE

SYN → (client asks to connect)

SYN/ACK ← (server agrees)

ACK → (client confirms — connection established)

SYN scan (-sS) sends SYN, receives SYN/ACK, then sends RST to cancel — never completes the handshake. This makes it stealthy (not logged) and faster than a full TCP connect scan.

7. COMMON SCAN EXAMPLES

```
Ping sweep: nmap -sn 192.168.0.0/24
```

```
Quick scan: nmap -sV -T4 <IP>
```

```
Full port scan: nmap -sV -p- <IP>
```

```
Stealth scan: sudo nmap -sS -Pn <IP>
```

```
UDP top 20: sudo nmap -sU --top-ports 20 <IP>
```

```
Xmas scan: sudo nmap -sX -p 1-999 <IP>
```

```
OS + version + scripts: sudo nmap -A <IP>
```

```
Vuln scan: nmap --script=vuln <IP>
```

```
FTP anon check: nmap -p 21 --script=ftp-anon <IP>
```

```
Save output: nmap -oA scan_results <IP>
```